GreenGrid EV Charging
Curation Foundations Case Study

End-to-End Data Curation, Modeling, Metrics and Lineage

1. Executive Summary

GreenGrid operates electric-vehicle charging stations and collects information from charging stations, its mobile application, station partners and customer support. These systems use different identifiers, formats and status values. Data can be duplicated, incomplete, late or corrected after it was first received. If each team interprets the data differently, dashboards and operational decisions may disagree.

The proposed solution moves data through an end-to-end platform: source systems → ingestion → Bronze/raw → Silver/curated → Gold/consumer-ready datasets → data products and a shared semantic layer → business consumers. Data modeling organizes the information for different uses, metric definitions establish consistent calculations, and lineage records how data was produced. The result is more reliable, explainable and reusable information.

Important note: All IDs, sample rows, thresholds and calculations in this report are illustrative examples. GreenGrid must approve the final business rules, especially the definition of a successful charging session.

2. Business Problem and Objectives

GreenGrid needs a consistent view of charging activity, station performance and customer issues.

- Combine data from multiple source systems without losing the original source information.
- Identify duplicate records and standardize inconsistent names, timestamps, units and statuses.
- Resolve records that refer to the same real-world charging station.
- Handle late-arriving events and corrections using clear source-authority and version rules.
- Provide different consumers with data structured for their needs.
- Define shared business terms and metrics so reports calculate results consistently.
- Trace a report value back to the source records and transformations that produced it.

3. End-to-End Architecture and Flow

Order	Stage	Main purpose	Typical output
1	Source systems	Create operational data	Station events, bookings, partner records, support tickets
2	Ingestion	Move data into the platform and record delivery metadata	Batch files, API records, streaming events
3	Bronze / Raw	Preserve received data with minimal changes	Raw tables plus source and ingestion metadata
4	Silver / Curated	Validate, standardize, deduplicate and resolve entities	Trusted, consistent domain tables
5	Data modeling / Gold	Organize curated information for specific analytical needs	Domain models, dimensional tables, reporting datasets
6	Semantic layer and metrics	Define shared business meaning and calculation rules	Consistent terms, measures and reusable logic
7	Data products and consumers	Publish useful, governed data for teams and tools	Documented datasets, dashboards, applications
8	Lineage and monitoring	Track origin, transformations, quality and dependencies	Traceability, alerts and audit history

Lineage and monitoring are cross-cutting capabilities: they should capture information throughout the pipeline, not only at the final stage. A semantic layer may sit above curated or Gold datasets; it is a logical business definition layer, not necessarily a separate physical storage layer.

4. Source Systems and Their Data

Each source system is designed for a different operational purpose. Before ingestion, the platform team should identify the owner, schema, identifiers, update frequency, expected freshness, data quality risks and correction behavior.

Source system	Example data	Likely ingestion approach	Key considerations
Charging stations	Session ID, station ID/name, start/end time, energy, status	Streaming/events when near-real-time monitoring is required; otherwise scheduled loads	Duplicate events, event ordering, late status updates, unit and timestamp consistency
Mobile application	Booking ID, user ID, selected station, booking status	API or event ingestion; incremental extraction where appropriate	Booking changes, cancellations, user privacy, station-name variations
Station partners	Partner station ID, station name, city, partner name, location	Batch files or API/incremental ingestion, depending on partner capability	Different naming conventions, stale reference data, partner-specific IDs
Customer support	Ticket ID, user ID, session ID, issue type, resolution	API, event-based or scheduled ingestion	Tickets may arrive later than sessions; a ticket is evidence of an issue but not automatically proof of the final session outcome

These ingestion choices are recommendations for the example scenario, not fixed requirements. The final method depends on each source's capabilities and the freshness required by the business.

5. Ingestion: Bringing Data into the Platform

Ingestion transfers data from each source into the data platform. The platform should record metadata such as source system, source record or event ID when available, ingestion timestamp, source event timestamp, batch/run ID and schema version.

5.1 Batch and streaming

- Batch ingestion moves groups of records at scheduled times. It is suitable when data can arrive periodically, such as a partner's daily station file.
- Streaming or event-based ingestion processes individual events continuously or in small increments. It can support timely station monitoring.
- API and incremental ingestion retrieve changed or newly created records without repeatedly loading the entire source, when the source supports this.

5.2 Ingestion checks

- Check that the incoming file, message or API response can be read.
- Validate the expected schema and required fields.
- Capture rejected records and failure reasons instead of silently discarding them.
- Make retries safe so that reprocessing the same delivery does not create unintended duplicates.
- Monitor freshness, volume changes and ingestion failures.

6. Bronze Layer: Raw Data

The Bronze or raw layer stores the data as received, with minimal alteration. Its purpose is to preserve the original evidence, support replay and make it possible to investigate later corrections. Raw data can contain duplicates, inconsistent names and incomplete values; that is expected and should be made visible rather than silently hidden.

6.1 Illustrative raw records

Charging-Station Source

session_id	station_name	energy_kwh	session_status	start_time	partner_station_id
CS-204	GreenGrid MG Road	12	Successful	2026-10-12 10:00	P-890
CS-205	GreenGrid MG Road	0	Charging	2026-10-12 10:30	P-890
CS-204	GreenGrid MG Road	12	Successful	2026-10-12 10:00	P-890

- start_time — helps analyze session timing and duration.
- partner_station_id — links the charging station to the partner source.

Note: CS-204 appears twice with identical values. This duplicate can be identified and handled during curation.

2. Mobile Application Source

booking_id	user_id	station_name	booking_status	booking_time	payment_status
B-501	U-101	MG Rd Charging Point	Completed	2026-10-12 09:50	Paid
B-502	U-102	GreenGrid Whitefield	Confirmed	2026-10-12 10:20	Pending
B-503	U-101	MG Rd Charging Point	Cancelled	2026-10-12 11:00	Not applicable

- booking_time — helps analyze booking patterns and compare booking times with charging sessions.
- payment_status — helps track payment completion and identify pending payments.

3. Station-Partner Source

partner_station_id	station_name	city	partner_name	tariff_per_kwh	partner_status
P-890	Green Grid, MG Road	Bengaluru	Partner A	₹12	Active
P-891	GreenGrid Whitefield	Bengaluru	Partner B	₹14	Active
P-890	Green Grid, MG Road	Bengaluru	Partner A	₹12	Active

- tariff_per_kwh — records the rate charged per kWh.
- partner_status — indicates whether the partner relationship is active.

Note: P-890 appears twice. If partner_station_id is the unique identifier for a partner's station, this duplicate should be investigated before treating the table as a reliable reference.

4. Customer-Support Source

ticket_id	user_id	session_id	issue_type	ticket_status	created_at
T-701	U-101	CS-204	Charging interrupted	Open	2026-10-12 10:15
T-702	U-102	CS-205	Station unavailable	In progress	2026-10-12 10:35
T-703	U-101	CS-204	Refund request	Closed	2026-10-12 11:20

columns:

- ticket_status — helps track whether a customer issue is open, in progress, or closed.
- created_at — supports analysis of issue frequency and resolution timelines.

The repeated CS-204 and P-890 rows illustrate duplicate-looking records. In production, a repeated row should not be removed only because its values look identical: first determine whether it is a duplicate delivery, a legitimate repeated event, or a correction.

7. Silver Layer: Cleaned and Trusted Data

The Silver layer converts raw data into validated, standardized and linked records. Transformations should be documented, repeatable and governed by explicit rules.

7.1 Core curation activities

- Validation: check required identifiers, valid timestamps, allowed status values and reasonable energy values.
- Deduplication: use source event IDs, record keys, version information and source-specific rules to identify duplicate deliveries.
- Standardization: normalize station names, city values, timestamps, units and status vocabulary.
- Entity resolution: determine when different source records refer to the same real-world station, user or session.
- Missing and invalid data handling: flag or quarantine records according to rules; do not automatically replace missing energy with zero.
- Late data and corrections: use source event time, source version, authority and business rules to determine the valid state.
- Auditability: retain references to source records and record important transformations.

7.2 Entity resolution: matching the same station

The following names may refer to the same station: 'GreenGrid MG Road', 'MG Rd Charging Point' and 'Green Grid, MG Road'. String similarity can suggest a match, but a reliable match should use stronger evidence such as address, coordinates, partner ID or an approved station crosswalk. Ambiguous cases should be sent for review rather than automatically merged.

source_system	original_id	original_station_name	canonical_station_id
Charging station	ST-101	GreenGrid MG Road	GG-ST-001
Mobile app	APP-55	MG Rd Charging Point	GG-ST-001
Station partner	P-890	Green Grid, MG Road	GG-ST-001

The canonical ID is a stable platform identifier. The crosswalk preserves the relationship between each source's identifier and the canonical station, making matches explainable and reversible when necessary.

7.3 Example curated tables

Curated table	Illustrative fields	Purpose
silver_station	canonical_station_id, standard_name, city, partner_id	One governed record per recognized station
silver_session	session_id, canonical_station_id, start_time, end_time, energy_kwh, current_status, source_updated_at	Validated charging-session state
silver_booking	booking_id, user_id, canonical_station_id, booking_status, updated_at	Standardized app bookings
silver_support_ticket	ticket_id, user_id, session_id, issue_type, resolution_status, created_at	Support issues linked to relevant records
station_identity_map	source_system, original_id,	Explainable source-to-canonical entity

	canonical_station_id, match_method, confidence, reviewed_by	mapping

Table designs are illustrative. For example, a source may not provide an end time or a stable event ID; the model must reflect what the source actually supplies and document any limitations.

7.4 Handling a corrected charging session

Suppose session CS-204 is first reported as successful with 18 kWh, then a later validated correction says it was interrupted after delivering 12 kWh because of a station fault. The platform should not simply trust whichever row arrived last without checking source authority and version rules.

session_id	version	status	energy_kwh	interpretation
CS-204	1	Successful	18	Initial reported state
CS-204	2	Interrupted — station fault	12	Later validated correction

A current-state table can expose the latest valid state, while a history table preserves prior versions and the reason for change. Reports should be refreshed or restated according to the agreed correction policy.

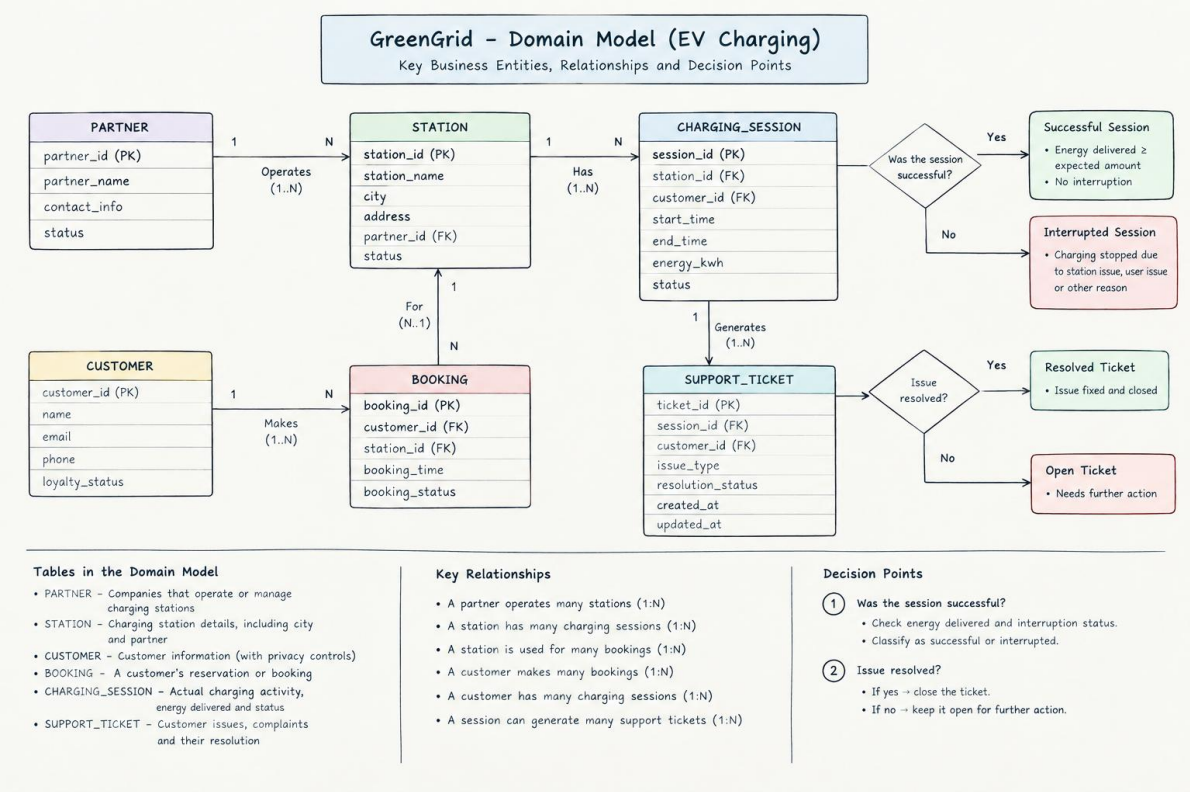
8. Data Modeling for Different Consumers

A domain model, a dimensional model and a product data model serve different purposes. They are complementary design approaches, not necessarily three sequential physical layers.

8.1 Domain model

The domain model describes the important business entities and how they relate. For GreenGrid, the domain includes Station, Charging Session, Customer, Booking, Partner and Support Ticket.

- A station can have many charging sessions.
- A customer can create many bookings and support tickets.
- A support ticket may reference a charging session when the relationship is known.
- A booking and a charging session are related business events, but they should not be treated as identical records.



8.2 Dimensional model

A dimensional model is useful for analytics and dashboards. A star schema places measurable events in a fact table and descriptive attributes in dimension tables.

Table	Illustrative columns	Role
Fact_Charging_Session	session_id, station_key, date_key, customer_key, energy_kwh, duration_minutes, success_flag	Measures and keys for session-level analysis
Dim_Station	station_key, canonical_station_id, station_name, city, partner	Allows analysis by station, city and partner
Dim_Date	date_key, date, month, quarter, year	Allows time-based analysis
Dim_Customer	customer_key, approved customer attributes	Allows permitted customer-level analysis

The fact table's grain should be explicitly defined—for example, one row per finalized charging session. If the business also needs attempt-level or event-level analysis, those may require separate fact tables. Customer attributes should be minimized and access-controlled.

8.3 Product data model

A Charging Session Data Product packages a trusted dataset for reuse. It includes more than table columns: it should also have an owner, description, schema, refresh expectation, quality checks, access rules, support contact and versioning policy.

Field	Meaning
session_id	Stable session identifier
canonical_station_id	Resolved station identifier
start_time / end_time	Session timing with defined timezone handling
energy_kwh	Validated energy delivered in kilowatt-hours
current_status	Approved status vocabulary

success_classification	Business-rule result for finalized sessions
updated_at	Timestamp representing the valid update under the chosen rule

9. Gold / Consumer-Ready Data/Curated Layer

Gold is where curated data is shaped for specific reporting or business needs. Depending on the architecture, it may contain dimensional tables, aggregates, reporting views or datasets built from curated domain tables. Not every organization needs a separate Gold storage layer.

Consumer	Likely data view	Questions it helps answer
Operations	Current station and session status, station-level quality indicators	Which stations have active faults? Which sessions need attention?
Customer support	Session, booking and support-ticket view	What happened during this customer's session, and is the information current?
Business analysts	Dimensional model and metric-ready reporting dataset	How do energy delivered and success rate vary by city, station, partner and month?
Customer-facing application	Approved station availability and status service/data product	Which stations are available, based on the latest reliable information?

10. Semantic Layer: Shared Business Meaning

A semantic layer gives business terms a common meaning across reports, dashboards and analytics tools. It defines or exposes entities, relationships, dimensions and measures so each team does not independently invent the same logic.

Business concept	Shared definition example
Charging session	A uniquely identified charging event, with explicit rules for attempts, retries and session boundaries
Energy delivered	Validated energy delivered by a session, measured in kWh
Session city	The standardized city associated with the canonical station for that session
Successful charging session	A finalized session that meets the business-approved success criteria
Session success rate	Successful eligible finalized sessions divided by all eligible finalized sessions

The exact meaning of 'successful' must be approved by business and operations. For example, GreenGrid must decide whether a session is successful when charging begins, when a minimum energy target is reached, or when the session completes without a confirmed fault. Once approved, the rule should be implemented consistently and versioned.

11. Business Metric: Session Success Rate

11.1 Illustrative metric specification

Metric element	Proposed rule (requires GreenGrid approval)
Name	Charging Session Success Rate
Purpose	Measure the proportion of eligible finalized sessions that meet the approved success criteria
Numerator	Count of eligible finalized sessions classified as successful

Denominator	Count of all eligible finalized sessions, including finalized unsuccessful/interrupted sessions
Excluded records	Pending or otherwise non-finalized sessions; test or invalid records according to approved rules
Time attribution	Use an agreed timestamp, such as session end time, with a defined timezone
Corrections	Recalculate affected periods or publish restatements according to the correction policy
Owner	Business/operations metric owner, with a data owner responsible for implementation

11.2 Example calculation

session_id	status	eligible for metric?	successful?
CS-204	Successful	Yes	Yes
CS-205	Interrupted	Yes	No
CS-206	Successful	Yes	Yes
CS-207	Pending	No	Not yet classified

Under these illustrative rules:

Session Success Rate = (2 ÷ 3) × 100 = 66.67%

CS-205 remains in the denominator because it is finalized and unsuccessful. CS-207 is excluded because it is still pending. If CS-204 is later corrected to interrupted, the metric must be recalculated according to the approved correction policy. This example illustrates the calculation only; the eligibility and success rules must be confirmed by GreenGrid.

12. Transformation Rules and Data Quality

Issue	Example	Treatment
Duplicate delivery	The same source event arrives twice	Identify using source event ID/version and safe reprocessing rules
Inconsistent station name	GreenGrid MG Road / Green Grid, MG Road	Map to canonical station only when evidence supports the match
Different units	Energy supplied in Wh by one source and kWh by another	Convert to a common unit using documented conversion logic
Different status wording	Interrupted / Fault / Failed	Map only where meanings align; preserve source status for audit
Missing energy	energy_kwh is null	Flag as missing or unresolved; do not silently convert to zero
Impossible value	Negative energy without a valid business meaning	Validate, quarantine or investigate under source-specific rules
Late correction	Session status changes after initial reporting	Apply version/authority rules and update current state while retaining history

Quality rules should be measurable. Examples include percentage of sessions with valid station IDs, duplicate-event rate, percentage of unmatched stations, freshness delay and number of quarantined records. Thresholds and owners should be agreed with the business.

13. Data Lineage, Monitoring and Governance

Lineage records where data came from, how it changed and which downstream products use it. It supports troubleshooting, audits, impact analysis and explanations when a dashboard value changes.

Lineage point	Information to capture
Source	Source system, source record/event ID, source timestamp and schema/version where available
Ingestion	Ingestion timestamp, batch/run ID, delivery status and rejected-record reason
Curation	Validation results, deduplication rule, entity match/crosswalk, transformation version and correction history
Model/reporting	Curated tables used, join logic, metric-definition version, aggregation and reporting period
Published product	Product version, owner, refresh time, quality status and downstream consumers

Example: if the monthly success rate changes, lineage should help the team determine whether new source events arrived, a session was corrected, a station match changed, or the success-rate definition was updated. Monitoring should alert the responsible team when freshness or quality falls outside agreed limits.

14. Recommended Roles and Responsibilities

Role	Responsibility
Source-system owner	Explain source fields, identifiers, event meaning and correction behavior
Data engineer	Build ingestion, validation, transformations, retry handling and monitoring
Data steward / domain owner	Approve canonical entity mappings and resolve ambiguous matches
Business / operations owner	Approve definitions such as a successful charging session and metric exclusions
Analytics engineer / data modeler	Design curated and dimensional models and implement shared metric logic
Data product owner	Document the product, manage quality expectations, access and consumer feedback

15. Final End-to-End Summary

1. Source systems generate charging, booking, partner and support data.
2. Ingestion transfers the records and captures delivery metadata and errors.
3. Bronze preserves the received data so the original evidence remains available.
4. Silver validates and standardizes the data, removes confirmed duplicate deliveries, resolves entities and manages corrections.
5. Domain, dimensional and product models organize the trusted data for different purposes.
6. Gold or other consumer-ready datasets expose the appropriate shape for reporting and operations.
7. The semantic layer defines shared business meanings; approved metric rules calculate measures consistently.
8. Data products publish documented and governed information to Operations, Customer Support, Business Analytics and applications.
9. Lineage and monitoring track the data path, quality, transformations, corrections and downstream impact throughout the process.

16. Curation Foundations Review Checklist

Learning outcome	How this case study demonstrates it

Data modeling	Domain model for business entities; dimensional star schema for analytics; product model for reusable delivery
Semantic layer	Shared definitions for charging session, energy delivered, city and successful session
Entity resolution	Canonical station ID and source-to-canonical mapping table
Business metrics	Success-rate numerator, denominator, exclusions, timestamp and correction policy
Transformation logic	Validation, deduplication, standardization, unit conversion and late-correction handling
Lineage	Source-to-report traceability with transformation versions and dependencies

17. Conclusion

GreenGrid's curation approach turns fragmented operational records into trusted and reusable data. The most important principle is to keep the flow explicit: ingest the records, preserve raw evidence, curate and resolve the data, model it for the intended use, establish shared business meaning, calculate metrics using approved rules, and retain lineage throughout. Curation is not just cleaning data; it is making data reliable, understandable, traceable and fit for its consumers.